

A WiFi Fingerprint Augmentation Method for 3-D Crowdsourced Indoor Positioning Systems

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Abstract—WiFi received signal strength (RSS)-based fingerprinting has attracted much attention in indoor positioning in the past decade. One WiFi fingerprint comprises multiple RSS values annotated with the location (reference point) where the signals are obtained. The positioning accuracy of WiFi RSS fingerprinting-based indoor positioning systems is highly reliant on the data volume of the observed signals. However, taking such data in a large complex indoor area is usually time-consuming and labor-intensive. In recent years, crowdsourcing approaches have been proposed to collect WiFi data and record the location by utilizing the trajectories of common users to reduce the burden of constructing the database. Nevertheless, crowdsourced data is usually sensitive to crowd density. The data coverage is usually not enough to cover the entire targeted environment to provide good positioning accuracy, particularly at the beginning stage of constructing a database. Besides, it is also expected that some regions do not have enough fingerprints to provide good positioning performance since they do not have as many visitors as others. Therefore, this paper proposes a WiFi fingerprint augmentation method to generate more fingerprints by predicting RSS values on unsurveyed locations through a multivariate Gaussian process regression (MGPR) model. Evaluations are conducted on an open-source crowdsourced WiFi fingerprint dataset collected in an actual multi-floor university building. The experiment results show that the proposed WiFi fingerprint augmentation method can enhance the global data coverage (considering the entire building) to reduce the positioning error by 5% to 20%. Also, the proposed method can sharply reduce the positioning error in some indoor regions by improving local data density (considering a 2D region on a certain floor).

Index Terms—Indoor positioning, WiFi fingerprinting, machine learning, augmentation

I. INTRODUCTION

Indoor positioning has attracted tremendous research interest in recent years. Although GPS has been served for years to provide location-based services in outdoor scenarios, it cannot provide highly reliable positioning accuracy in indoor environments due to signal attenuation and multi-path effects. Therefore, researchers have developed a variety of positioning techniques, such as using magnetic field [1]–[3], radio frequency identification (RFID) [4]–[6], and WiFi [7]–[9], to compensate for the shortage of GPS in indoor positioning.

Nowadays, most mobile communication devices, such as mobile phones, smartwatches, and laptops, are embedded with WiFi chips to detect WiFi signals in indoor environments.

WiFi is widely adopted in many public and private areas to provide users with internet connections. Since no extra infrastructure is needed, fingerprinting based on WiFi received signal strength (RSS) has become the most prevailing method to solve the indoor positioning problem over the last decade [10]. This fingerprinting technique usually has two stages: online and offline stages. In the offline stage, a radio map is constructed by multiple WiFi RSS fingerprints that are annotated with the location where the signal was acquired. In the online stage, the user's test RSS value is compared to the fingerprints on the radio map to determine the most likely location where the signal was acquired.

A critical factor that affects the positioning accuracy using WiFi fingerprinting is the number of fingerprints on the radio map. Volunteers are asked to collect WiFi data at hundreds of reference points (RPs) in the investigated indoor region to get sufficient fingerprints. Such a site survey is usually time-consuming and labor-intensive [11]. The study from Wang *et al.* [13] mentions that it took about 10 hours for two graduate students to take fingerprints at 150RPs of an area of $281m^2$. It is even worse when considering a multi-floor/building scenario, such as an airport or shopping mall.

Therefore, a crowdsourcing approach has been proposed to collect WiFi data and record the location by utilizing the trajectories of common users. With the inertial sensors and WiFi card equipped in users' smartphones, the WiFi RSS fingerprints can be collected and annotated by their location along with their movements [12]. However, relying on crowdsourced data also brings some challenges. For example, it is expected that the data amount will be very low at the beginning stage of deploying a crowdsourced indoor positioning system in a certain building due to a limited number of users. Such data is not enough to cover the entire indoor environment to provide an accurate positioning service. Besides, the crowdsourced data cannot guarantee to be updated everywhere as time changes though more user contributions are involved as the user number grows. This attributes to low positioning accuracy in some regions as they do not have as many visitors as others.

This paper proposes a machine learning-based WiFi fingerprint augmentation method for 3-D crowdsourced indoor positioning systems in large, complex indoor environments. A multivariate Gaussian process regression (MGPR) model is

proposed to estimate the joint distribution of all training WiFi signals in a certain building. We use such a model to predict RSS values on the virtual RPs interpolated from the training RPs to generate more WiFi fingerprints. The contributions are summarized as follows:

- We propose a WiFi fingerprint augmentation method for 3-D crowdsourced indoor positioning systems. The proposed method can generate more WiFi fingerprints on unsurveyed locations to improve positioning accuracy.
- A multivariate Gaussian process regression (MGPR)-based Spatial WiFi Signal Modeling algorithm is designed to predict RSS values of each AP on the virtual RPs interpolated from the training RPs in a 3-D environment.
- Experiments in two scenarios of building-based augmentation and region-based augmentation are considered in this paper to comprehensively evaluate the improvement in positioning accuracy by the proposed method.

II. PROBLEM STATEMENT AND MOTIVATIONS

He *et al.* [14] introduces a technique for creating a radio map with fingerprints based on the radio propagation model. However, the radio model can be easily influenced by environmental changes, and the location of each AP is needed. Similarly, Jun *et al.* [15] presents an indoor positioning system named AP-Sequence that aims to reduce the human effort in fingerprint map creation with the location of each AP as a prior. Rather than using the propagation model, Sinhua *et al.* [16], [17] propose to use linear interpolation to augment the WiFi RSS to generate more training data. Besides the linear interpolation method, Sun *et al.* [19] proposed to use Gaussian process regression (GPR) to model the spatial distribution of the RSS of each AP, and predict the RSS values of each AP on unsurveyed locations to generate new fingerprints. Nevertheless, such a method needs to create multiple GPR models for all APs in the target area. Since the signal distribution of each AP is different in the complex indoor environment, tuning all GPR models will be inefficient and inappropriate for large indoor spaces with a large amount of APs.

Through the literature analysis, one problem of the existing augmentation methods is scalability. Either using linear interpolation or pass loss model (propagation model) cannot well predict the signal distribution in different indoor environments. The recent machine learning-based approaches use regression models to model each AP's signal distribution in the targeted indoor area. Although such methods have better generalization ability, they are not scalable and rational in practice. Usually, a WiFi RSS fingerprint dataset for a building with multiple floors will contain hundreds of APs. For example, an open-source dataset published in the EU Zenodo repository [20] contains more than 900 different MAC addresses in a five-floor building. Building and tuning the regression model for each AP for the entire building is highly time-consuming. Due to lack of scalability, all the aforementioned methods were designed and evaluated in small 2-D environments, such as a lab, an office, or a corridor associated with multiple rooms.

It is doubtful whether such methods could still maintain the performance considering a large complex environment.

Therefore, this paper proposes a novel data augmentation method by modeling the distribution of all WiFi fingerprints in the targeted building and augmenting them to improve the positioning accuracy of 3-D crowdsourced indoor positioning systems.

III. METHODOLOGY

This section presents the two major elements of the proposed WiFi fingerprint augmentation approach: Virtual Reference Point Generation (VRPG) and Spatial WiFi Signal Modeling (SWSM).

A. Virtual Reference Point Generation

As previously discussed, the proposed method aims to augment the WiFi fingerprint dataset by generating more WiFi fingerprints. For WiFi fingerprinting problems, each fingerprint is composed of a RSS vector $\mathbf{f} = [RSS_1, RSS_2, \dots, RSS_M]$ (M is the total number of detected APs in the entire dataset) and a location vector $\mathbf{c} = [x, y, z]$. Considering a 3D coordinate system, x , y and z stand for the coordinate along x , y and z -axis (in meters). To get more WiFi fingerprints for positioning, we first stipulate a strategy to determine the locations of where they should be generated. In this study, we design an interpolation approach.

For a targeted region on a certain floor at h m, such area can be described as a rectangle restrained by the maximum and minimum value of the coordinates of all the RPs along x and y -axes on the same floor. So that the four vertexes of such rectangle region are (x_{min}, y_{min}, h) , (x_{max}, y_{min}, h) , (x_{min}, y_{max}, h) and (x_{max}, y_{max}, h) .

We define a virtual RP in the targeted region as $\mathbf{c}^* = [x_i^*, y_j^*, h]$. The entire region is partitioned into multiple $1 \times 1 \text{m}^2$ grids along x and y axes. The coordinates of the virtual RPs are represented by the locations of the vertexes of each grid. So that:

$$i = x_{min}, x_{min} + 1, x_{min} + 2, \dots, x_{max} \quad (1)$$

$$j = y_{min}, y_{min} + 1, y_{min} + 2, \dots, y_{max} \quad (2)$$

hence, we can obtain t virtual RPs, where t can be calculated by:

$$t = (x_{max} - x_{min} + 1)(y_{max} - y_{min} + 1) \quad (3)$$

B. Spatial WiFi Signal Modeling

In this subsection, we present how we model the distributions of the spatial WiFi signal in a 3D environment to estimate the RSS values on the generated virtual RPs. For all RSS vectors $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_N\}$ in $\mathbf{F}$, and $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_N\}$ in $\mathbf{C}$, the mapping between the locations and the RSS vectors can be expressed by:

$$\mathbf{F} = \lambda(\mathbf{C}) + \eta \quad (4)$$

where η represents the independent and identically distributed Gaussian noise with zero mean and variance, which can be denoted by $\eta \sim \mathcal{N}(0, \sigma_n^2)$.

To solve the mapping problem, we assume that all RSS vectors in the investigated indoor area obey a multivariate Gaussian process of multiple high-dimensional joint Gaussian distributions. So that such a Gaussian process can be represented by the mean function $m(\mathbf{C})$ and covariance function $k(\mathbf{c}_i, \mathbf{c}_j)$, as shown in the following equation:

$$\lambda(\mathbf{C}) \sim GP(m(\mathbf{C}), k(\mathbf{C})) \quad (5)$$

$$m(\mathbf{C}) = \mathbb{E}[f(\mathbf{C})] \quad (6)$$

$$k(\mathbf{c}_i, \mathbf{c}_j) = \mathbb{E}[(f(\mathbf{c}_i) - m(\mathbf{c}_i))(f(\mathbf{c}_j) - m(\mathbf{c}_j))] \quad (7)$$

where $\mathbb{E}(\cdot)$ denotes the expectation operator. So that the covariance matrix $\mathbf{K}$ can be expressed by:

$$\mathbf{K} = \begin{bmatrix} k(\mathbf{c}_1, \mathbf{c}_1) & k(\mathbf{c}_1, \mathbf{c}_2) & \cdots & k(\mathbf{c}_1, \mathbf{c}_N) \\ k(\mathbf{c}_2, \mathbf{c}_1) & k(\mathbf{c}_2, \mathbf{c}_2) & \cdots & k(\mathbf{c}_2, \mathbf{c}_N) \\ \vdots & \vdots & \ddots & \vdots \\ k(\mathbf{c}_N, \mathbf{c}_1) & k(\mathbf{c}_N, \mathbf{c}_2) & \cdots & k(\mathbf{c}_N, \mathbf{c}_N) \end{bmatrix} \quad (8)$$

where N denotes the total number of WiFi fingerprints (or the total number of RPs).

Through the grid-based RP algorithm mentioned in the previous subsection, we will obtain N^* potential RPs $\mathbf{C}^*$. This means that we have N^* new WiFi RSS vectors (fingerprints) $\mathbf{F}^*$ to be inferred. The RSS vectors $\mathbf{F}$ in the original dataset and RSS vectors $\mathbf{F}^*$ to be inferred should also follow a joint multivariate Gaussian distribution, which can be stated by the following equation:

$$\begin{bmatrix} \mathbf{F} \\ \mathbf{F}^* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(\mathbf{C}) \\ m(\mathbf{C}^*) \end{bmatrix}, \begin{bmatrix} \mathbf{K}(\mathbf{C}, \mathbf{C})_{N \times N} & \mathbf{K}(\mathbf{C}, \mathbf{C}^*)_{N \times N^*} \\ \mathbf{K}(\mathbf{C}^*, \mathbf{C})_{N^* \times N} & \mathbf{K}(\mathbf{C}^*, \mathbf{C}^*)_{N^* \times N^*} \end{bmatrix} \right) \quad (9)$$

The posterior distribution $p(\mathbf{F}^*|\mathbf{F})$ can be then expressed as:

$$\mathbf{f}^*|\mathbf{f} \sim \mathcal{N}(\mathbf{K}(\mathbf{C}^*, \mathbf{C})\mathbf{K}(\mathbf{C}^*, \mathbf{C})^{-1}\mathbf{F}), \quad (10)$$

$$\mathbf{K}(\mathbf{C}^*, \mathbf{C}^*) - \mathbf{K}(\mathbf{C}^*, \mathbf{C})\mathbf{K}(\mathbf{C}^*, \mathbf{C})^{-1}\mathbf{K}(\mathbf{C}, \mathbf{C}^*)$$

Hence, the posterior mean and covariance of the observed RSS vectors can be computed to obtain a model to predict the new RSS vectors on the unsurveyed potential RPs.

In addition, the covariance function $k(\mathbf{c}_i, \mathbf{c}_j)$ (also called kernel function) plays a vital role in a Gaussian process to denote the relation between the RSS vectors and the corresponding RPs. One popular kernel function is the squared exponential kernel, which assumes that the process of the system is very smooth [21]. This is not suitable to describe the relationship among the high-dimensional RSS vectors in the large complex indoor scenario in our case. In [19], a mixture of Matern and Rational Quadratic (RQ) kernels performs the

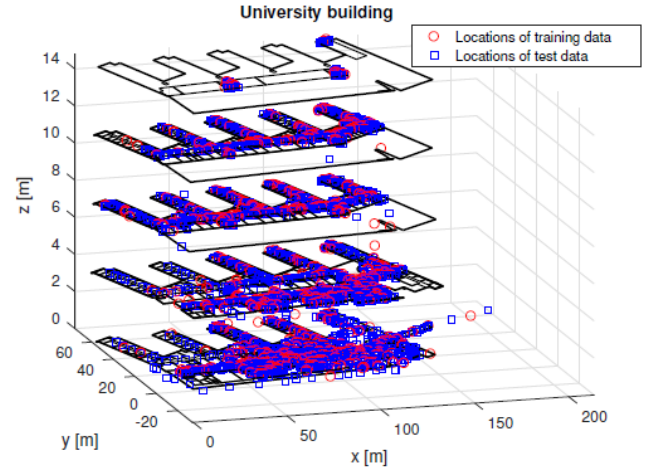


Fig. 1. 3D View of the dataset [20].

best in capturing the variation of RSS values in comparison with other kernels. The Matern kernel is defined as:

$$k_{\text{Matern}}(\mathbf{c}_i, \mathbf{c}_j) = \frac{1}{\Gamma(\epsilon)2^{\epsilon-1}} \left(\frac{\sqrt{2\epsilon}}{l} d(\mathbf{c}_i, \mathbf{c}_j) \right)^{\epsilon} K_{\epsilon} \left(\frac{\sqrt{2\epsilon}}{l} d(\mathbf{c}_i, \mathbf{c}_j) \right) \quad (11)$$

where ϵ denotes the smoothness of the function; l is the length scale; $K_{\epsilon}(\cdot)$ and $\Gamma(\cdot)$ are the gamma function and the modified Bessel function [21], respectively; d is the Euclidean distance which can be calculated by:

$$d = \sqrt{(\mathbf{c}_i - \mathbf{c}_j)^T (\mathbf{c}_i - \mathbf{c}_j)} \quad (12)$$

And the Rational Quadratic kernel can be described as:

$$k_{\text{RQ}}(\mathbf{c}_i, \mathbf{c}_j) = \left(1 + \frac{d(\mathbf{c}_i, \mathbf{c}_j)^2}{2\alpha l^2} \right)^{-\alpha} \quad (13)$$

where α stands for the shape parameter. So that the mixed kernel in our case is designed as follows:

$$k_{\text{mixed}} = \mu * k_{\text{Matern}} + \nu * k_{\text{RQ}} \quad (14)$$

where μ and ν are the weighting parameters. All the above mentioned hyperparameters can be optimized through minimizing the negative logarithmic marginal likelihood.

IV. EVALUATION

A. Preliminaries

1) *Dataset*: We evaluate the proposed WiFi fingerprint augmentation method on a long-term public benchmark open-source WiFi fingerprinting dataset [20]. The data was collected from a five-floor University building (about 22,570m²) in Tampere, Finland. The height of each floor is 0, 3.7, 7.4, 11.1 and 14.8m, respectively. 4648 fingerprints were collected, where 15% (697 fingerprints) were obtained for training and 85% (3951 fingerprints) for testing. A total of 992 MAC addresses (APs) were detected. The non-detected APs in each fingerprint are set to +100dBm.

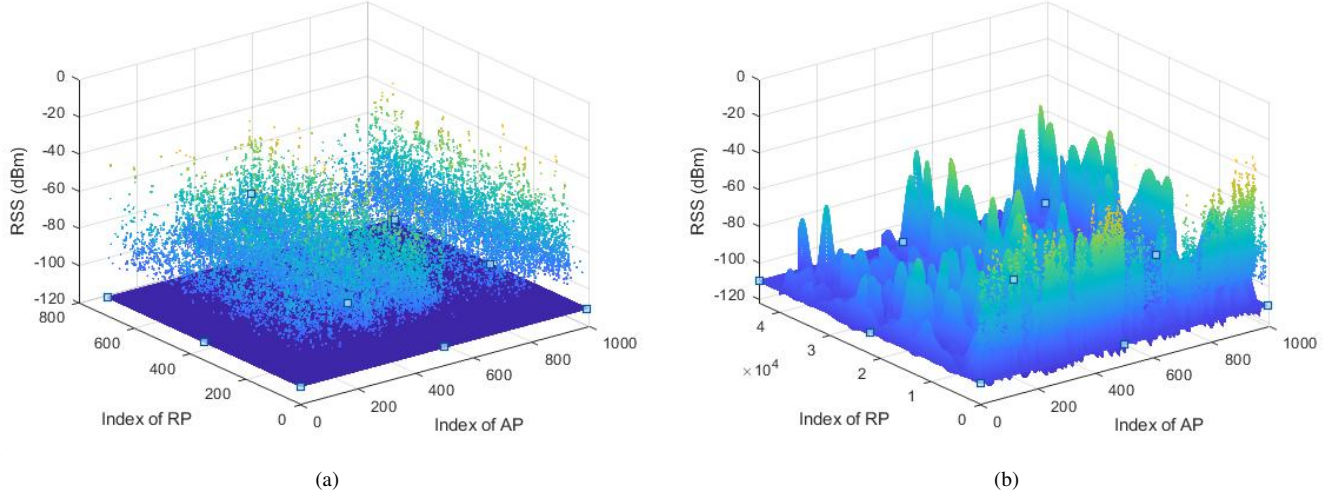


Fig. 2. Visualization of the all WiFi signals in the targeted building: (a) raw training data (the non-detected signal value (100) is replaced by -110); (b) augmented training data.

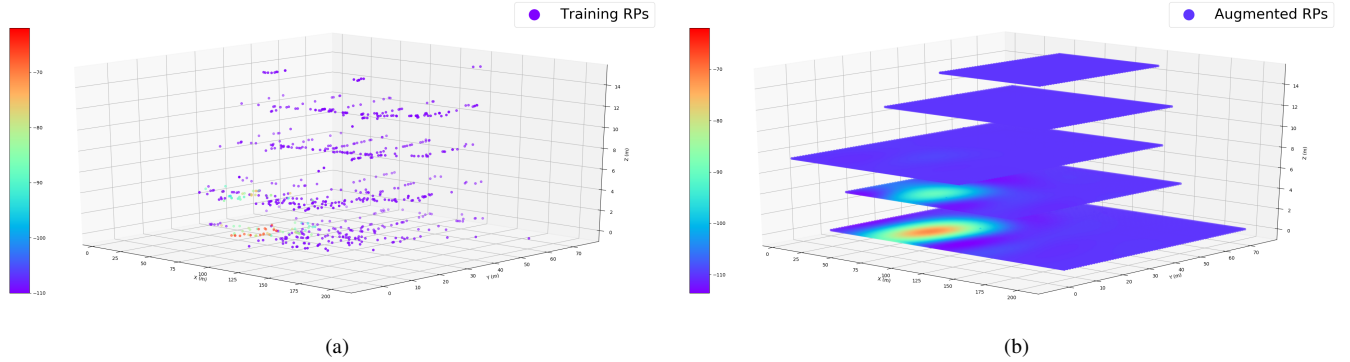


Fig. 3. 3-D Visualization of the WiFi signal distribution of AP 250 in the targeted building: (a) raw training data (the non-detected signal value (100) is replaced by -110); (b) augmented training data.

2) *Data pre-processing*: The RSS values from non-detected APs were set to -110dBm before any analysis and experiments in both training and test sets. All RSS values were normalized by the maximum and minimum values before training the MGPR model. The 3-D coordinates were also normalized by the maximum and minimum values along each training axes. The equation for normalization is shown below:

$$m_i^{norm} = \frac{m_i - m_{\min}}{m_{\max} - m_{\min}} \quad (15)$$

where one sample m_i is normalized to m_i^{norm} by the maximum value $m_{\max}$ and minimum value $m_{\min}$.

3) *Positioning algorithm*: This study did not propose a new positioning algorithm but applied a widely adopted Weighted K-Nearest Neighbors algorithm (WKNN) to predict the locations. WKNN aims to find the k fingerprints in training set with the nearest weighted distance to the test sample. The WKNN algorithm in this study was implemented using Scikit-learn [22]. The distance metric was set to Euclidean distance, and k was 3.

B. Spatial WiFi Signal Modeling

We first trained the proposed MGPR model with the pre-processed training data. Then, we generated new fingerprints by predicting WiFi RSS values on the virtual RPs at the interval of 1m with the well-trained MGPR model. The new and raw fingerprints were then mixed as augmented training data.

Figure 2 visualizes the WiFi signals against the index of RP and fingerprint. It can be seen from Figure 2a that the raw training data is highly sparse. The sparse effective RSS values (not -110dBm) associated with limited training RPs are not enough to provide smooth distribution of the WiFi signals for positioning use. Differently, the augmented dataset has a large number of effective RSS values predicted by the MGPR model on virtual RPs, which can provide more information when estimating the location of a given test fingerprint.

In addition, Figure 3 gives an example of the comparison between the WiFi signal distribution of one single AP (No.250) in the 3-D environment. It is obvious that the augmented data

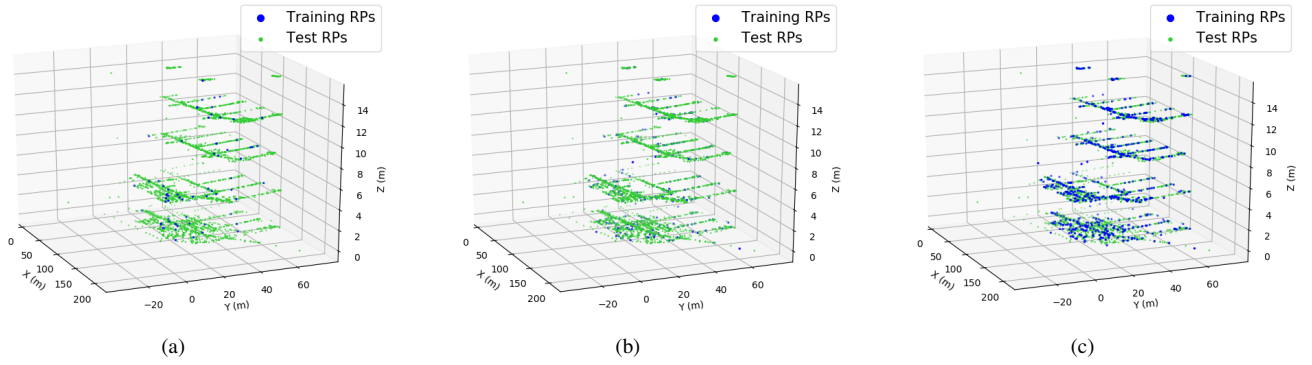


Fig. 4. 3D-visualization of the partial training data and the test data: (a) sampling ratio = 10%; (b) sampling ratio = 50%; (c) sampling ratio = 100%.

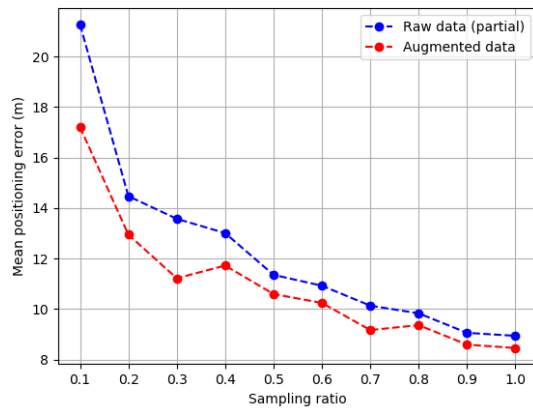


Fig. 5. Mean positioning error of using different percentage of the training set.

is able to provide more effective RSS values than the raw training data.

Therefore, this informative augmented dataset is expected to provide more accurate positioning accuracy, which will be discussed in the next subsection.

C. Building-based augmentation

1) *Experimental settings*: In this experiment, we randomly selected a portion of the training data from the original training set with different percentages (from 10% to 100%). This was to simulate the growth of the training data coverage in the entire building. As the crowdsourced indoor positioning systems employ users' contributions to construct the training set, the training data amount and its coverage are expected to be very limited at the beginning stage due to few users. As time goes by, the training set is expected to grow and finally cover the entire building as more and more users and their contributions are involved.

It can be seen from Figure 4 that the coverage of the partial training data is very limited when the sampling ratio is very low. Most of the areas are not surveyed. As the sampling ratio

increases, the training data grows and finally covers the entire building.

We trained the proposed MPGR model with such partial training data and predicted new RSS values on the interpolated RPs to generate new fingerprints. The new fingerprints and the partial training fingerprints were then mixed into a new augmented dataset. Positioning performance evaluations were then conducted based on the raw and augmented training sets.

2) *Evaluations and analysis*: As shown in Figure 5, the positioning error decreases sharply at the beginning as the sampling ratio increases from 10% to 20%. Then, it declines steadily as more training fingerprints are involved. The positioning error with the augmented dataset shows a similar trend as the partial training data but less error at each stage. We can observe from the figure that although there is a slight fluctuation, augmented data can always provide better positioning accuracy than the raw data. It can be seen from Table I that the augmented data can provide about 5% to 20% improvement in mean positioning accuracy.

We can also observe that the augmented data showed significant improvement (10-20%) at the beginning stage when only 40% of raw training data were used. However, the improvement decreased steadily when more raw training data were involved. This is because of the saturation of raw training data. Although the new fingerprints generated by the MGPR model can complement the limited number of raw training data at the beginning, the contribution of such new fingerprints weakens as the coverage of the raw training data grows.

Also, we can see some fluctuations in the improvement of positioning accuracy. This is because the distribution of raw training data is usually uneven, which attributes to different positioning accuracy in different areas. Such a situation is consistent with the real situation that more users may visit some areas in the building, which provides more informative data and higher positioning accuracy. This motivates us to conduct the following experiment on region-based augmentation.

D. Region-based augmentation

1) *Experimental settings*: As previously discussed, in crowdsourced indoor positioning systems, the data amount and coverage are always a key to the positioning error. This

TABLE I
POSITIONING PERFORMANCE EVALUATIONS ON BUILDING-BASED AUGMENTATION

Sampling ratio	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Positioning error with raw data (m)	21.27	14.47	13.57	13.01	11.36	10.93	10.14	9.83	9.06	8.94
Positioning error with augmented data (m)	17.21	12.95	11.22	11.72	10.60	10.24	9.17	9.36	8.59	8.46
*Improvement in meter (m)	4.06	1.53	2.36	1.29	0.76	0.68	0.97	0.47	0.48	0.48
†Improvement by percentage (%)	20.07	10.54	17.35	9.88	6.68	6.23	9.54	4.77	5.16	5.40

* Improvement in meter = positioning error with augmented data - positioning error with raw data

† Improvement by percentage = Improvement in meter/ positioning error with raw data *100%

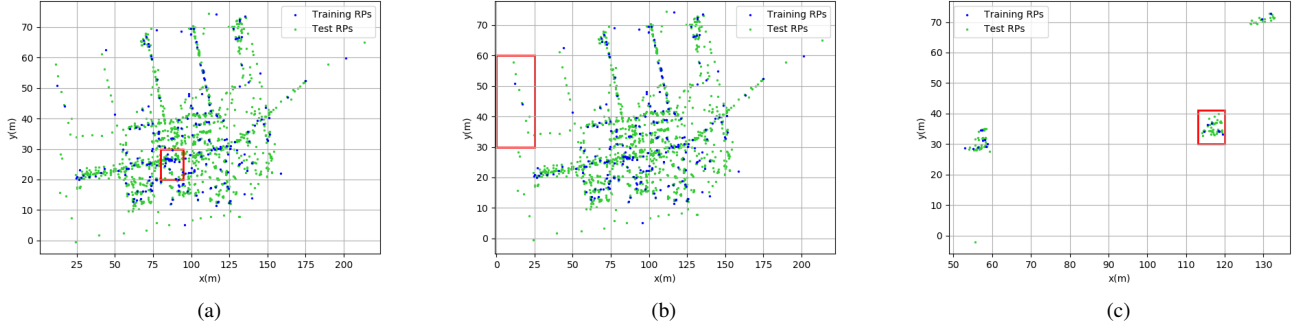


Fig. 6. 2D-visualization of the selected region (in red box): (a) region 0; (b) region 1; (c) region 2.

TABLE II
DETAILS OF THE SELECTED REGIONS

	Region 0	Region 1	Region 2
Range (m)	(85, 20) (95, 30)	(0,30) (25, 60)	(113, 30) (120, 41)
Floor (m)	0 (ground-floor)	0 (ground-floor)	14.8 (fourth-floor)
Area (m^2)	100	750	77
*Number of local training data	20	2	6
†Number of local test data	72	8	44
2-D Isolation factor	6.57	26.03	37.91
3-D Isolation factor	6.57	20.22	7.29
Regional information density	0.2	0.003	0.08

*This denotes the number of training samples in the target region in the raw training set.

† This denotes the number of test samples in the target region in the test set.

experiment was set to simulate the situation that users barely visit some indoor areas. We selected three different indoor regions representing different cases that we would have in real life. As shown in Figure 6, the three regions are rectangular shapes denoted by the coordinates of the lower-left corner and the upper-right corner. The details of the three regions are illustrated as follows:

- Region 0: this region is in the central area of the ground floor, which most people should visit. We selected this region to evaluate the generalization ability of the proposed model in region-based augmentation in general.

- Region 1: this region is in a corridor on the ground floor at the edge of the building. It is in the corner and away from the central area with a few visits. We selected this region to evaluate the generalization ability in the interference region with less relevance to other areas.
- Region 2: this region is on the top (fourth) floor and far away from any other areas on the same floor. This was set to evaluate the generalization ability in the isolated interference region with much less relevance to other areas.

For each region, we calculated several features and listed them in are presented in Table II. For better illustration and analysis of region-based augmentation, we defined the following new parameters to represent the features of the targeted regions:

- 2-D Isolation Factor (2DIF): This parameter describes the relationship between the targeted/inference region and other areas with training data. It is calculated by the physical distance between the centroid of the targeted region and the nearest sample outside the region from the training set on the same floor (consider 2-D coordinates).
- 3-D Isolation Factor (3DIF): This is similar to 2DIF but calculates the physical distance between the centroid of the targeted region and the nearest sample outside the region in the entire building (consider 3-D coordinates).
- Regional information density (RID): This parameter describes the information density of the targeted/inference region. It is the ratio between the number of local training samples and the region's area.

TABLE III
COMPARISON OF POSITIONING ERROR USING DIFFERENT DATA

	Region 0	Region 1	Region 2
Positioning error with raw data (PER) (m)	8.39	21.54	5.74
Positioning error without local training data (PEL) (m)	9.52	32.87	58.08
Positioning error with augmented data (PEA) (m)	7.43	30.74	18.00
*Improvement in meter (m)	0.96	2.13	40.08
†Improvement in percentage (%)	11.44	6.48	69.01

* Improvement in meter = PEL - PEA

† Improvement by percentage = Improvement in meter/ PEL *100%

In all cases in this experiment, the training samples in the targeted regions were removed from the training set to train the MGPR models. With the well-trained models, we calculated the positioning error using different settings of the training data for comparison:

- Positioning error with raw data (PER): using the entire raw training set.
- Positioning error without local training data (PEL): the samples in the targeted region were removed from the raw training set.
- Positioning error with augmented data (PEA): using the augmented data: newly generated data in the targeted area and the raw training data without the samples in the target area.

Since we evaluated local positioning errors in specific regions, only the test samples in the regions were kept.

2) *Evaluations and analysis*: We can observe from Table III that when there is no training data in the targeted regions, the proposed method can provide much lower PEA than PEL. The improvement is 11.44%, 6.48% and 69.01%, respectively. Particularly in region 2, the augmented data can provide almost 70% improvement in positioning accuracy. This is because region 2 is isolated and far away from any other areas in the same building. As listed in Table II that region 2 has a much higher 2DIF of 37.91m than others. This can also be found in Figure 6c that region 2 is about 30m away from the area in the upper-right corner and 55m away from the area on the left side. Therefore, the weighted nearest neighbors estimated by the WKNN algorithm are also far from region 2 when there is no local training data, which contributes to extremely low PEL. Although region 2 shows relatively lower 3DIF, this does not mean WKNN can find the nearest neighbors from the adjacent floor(s). This is because the spatial Euclidean distance used by WKNN cannot well describe the signal difference in 3-D environments. Nevertheless, benefitting from the lower 3DIF of region 2 (7.29m) than Region 1 (20.22m), the proposed MGPR model can learn more information from the adjacent floor(s) to predict new fingerprints with higher quality for positioning. That is also why region 2 shows more significant improvement than region 1.

In addition, region 0 shows both the lowest PEL (9.52m) and PEA (7.43m) among all regions. This is because this region is

more informative with the lowest 2DIF and 3DIF. Since region 0 is in the central area on the ground floor with comprehensive data coverage, the WKNN positioning algorithm can still find the nearest neighbors not far away from the region. Hence, the PEL is the lowest. Also, being more informative, the MGPR can provide a higher quality of new fingerprints in the region as the model can learn more from the training data surrounding the targeted region. So region 0 also shows the lowest PEA.

Besides, although region 0 has a higher RID (0.2) than region 2, region 2 shows a lower PER of 5.74m than region 0 (8.39m). This is because region 0 is in the central area with a lower 2DIF and 3DIF. The signals from the adjacent regions might lead to unnecessary ambiguity when comparing the test fingerprint to the training fingerprints. On the contrary, as region 2 has moderate RID but is isolated from other areas on the same floor, the positioning algorithm can predict accurate results with less interference from the raw training samples in other areas.

V. CONCLUSION

In this paper, we have proposed a WiFi fingerprint augmentation method for 3-D crowdsourced indoor positioning systems in large complex indoor environments. A key attribute of augmentation with the proposed method is the proposed SWSM algorithm that creates an MGPR model to estimate the joint distribution of all WiFi signals in a 3-D environment. We generate more fingerprints by predicting the RSS values with the MGPR model on virtual RPs interpolated by VRPG to augment the training set. Two scenarios of building-based augmentation and region-based augmentation have been considered and evaluated. The experiment results have shown that the proposed WiFi fingerprint augmentation method is able to improve the positioning accuracy with insufficient data coverage either globally (considering the entire building) or locally (considering a 2D region on a certain floor).

Our future work will design an automatic region partitioning algorithm associated with a WiFi fingerprint density evaluation algorithm to determine which region(s) should be augmented in a 3-D environment.

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